Conserve mechanical energy

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Track kinetic and potential energy when nonconservative work is negligible.

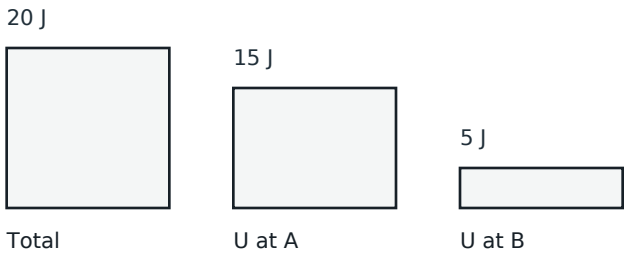
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 7.*
+150 J; the object gains gravitational potential energy.

02. Worked example: explain the model
U 50 to 0, K 0 to 50; object-Earth system, negligible drag, height zero at arrival.

03. Read and interpret the model
5 J and 15 J.

Model / diagram check



04. Apply the physics
20 J each.

05. Explain your reasoning
No; include thermal/deformation/sound energy and surroundings as appropriate.

06. Review: connect an earlier idea *Review: Unit 3, Day 12.*
Both become 4 times as large; p=mv and average force magnitude=change in momentum/time.

07. Exit check
18 J.

08. STRETCH: test the model
Same final speed for equal initial speeds; path length, time and acceleration histories can differ.

Conserve mechanical energy

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Track kinetic and potential energy when nonconservative work is negligible.

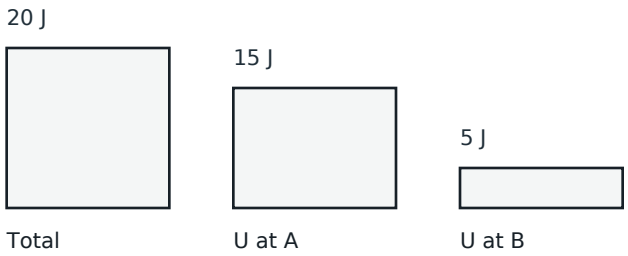
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 7.*
$\Delta U = -150$ J; gravity does +150 J work.

02. Worked example: explain the model
U 50 to 0, K 0 to 50; object-Earth system, negligible drag, height zero at arrival.

03. Read and interpret the model
5 J and 15 J.

Model / diagram check



04. Apply the physics
Initial E 100+16=116 J; final U 20, K 96; $v = \sqrt{96} = 9.80$ m/s.

05. Explain your reasoning
No; include thermal/deformation/sound energy and surroundings as appropriate.

06. Review: connect an earlier idea *Review: Unit 3, Day 12.*
A has 1/4 of B's speed and kinetic energy: $v = p/m$, $K = p^2/(2\,m)$.

07. Exit check
$v^2/(2\,g) = 5$ m.

08. STRETCH: test the model
Initial K=16 J; 3 m needs 60 J, so no. Maximum rise $v^2/(2\,g) = 0.8$ m.